Bias Mitigation Techniques

Overview of Bias Mitigation

1. Data-Level Interventions (Sampling/re-weighting)
2. Preprocessing: Modify dataset before training (Feature transformation/removal)
3. In-processing: Incorporate fairness during training (modified optimization function)
4. Post-processing: Adjust outputs after training (Threshold Adjustment, re-ranking)
5. Beyond metrics

Bias Mitigation Techniques

Data-Level Interventions

1. Balanced Sampling / Reweighting

Random oversampling (duplicating minority examples) or undersampling (removing majority examples) to achieve equal class proportions per training batch.

If the group population is sparse then **each sample from that group is given higher weight** resulting in debiasing the data.

Adding weights to sample i from group a_i as $w_i = 1/Pr(A = a_i)$.

The loss function is then modified to $\sum wi \cdot \ell(yi, \hat{y}i)$.

1. Balanced Sampling / Reweighting

Table 5 Sample job-application relation with weights.

Sex	Ethnicity	Highest Degree	Job Type	Cl.	Weight
m	native	h. school	board	+	0.75
m	native	univ.	board	+	0.75
m	native	h. school	board	+	0.75
m	non-nat.	h. school	healthcare	+	0.75
m	non-nat.	univ.	healthcare	-	2
f	non-nat.	univ.	education	-	0.67
f	native	h. school	education	-	0.67
f	native	none	healthcare	+	1.5
f	non-nat.	univ.	education	-	0.67
f	native	h. school	board	+	1.5

$$P_{exp}(Sex = f \mid x(Class) = +) = 0.5 \times 0.6$$

but its actual probability is 20%. *So the weight* W *will be:*

$$W(Sex = f \mid x(Class) = +) = \frac{0.5 \times 0.6}{0.2} = 1.5 \ .$$

Similarly the weights of the other combinations are:

$$W(Sex = f \mid x(Class) = -) = 0.67$$
$$W(Sex = m \mid x(Class) = +) = 0.75$$
$$W(Sex = m \mid x(Class) = -) = 2 \ .$$

Kamiran, Faisal, and Toon Calders. "Data preprocessing techniques for classification without discrimination." Knowledge and information systems (2012): 1-33

2. Datasheets for Datasets

Proposed by Timnit Gebru (co-authored the ‘Gender Shades’ paper) and colleagues in 2018.
A datasheet typically covers:

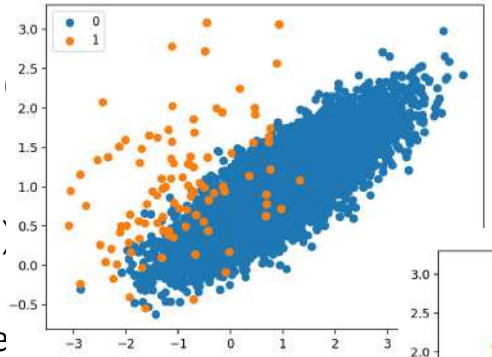
- Motivation: Why created, by whom, for what task.
- Composition: Size, features, demographics, errors or noise.
- Collection Process: Methods, time/place, incentives, consent.
- Recommended Uses and Limitations: Intended tasks, known biases, maintenance plans.

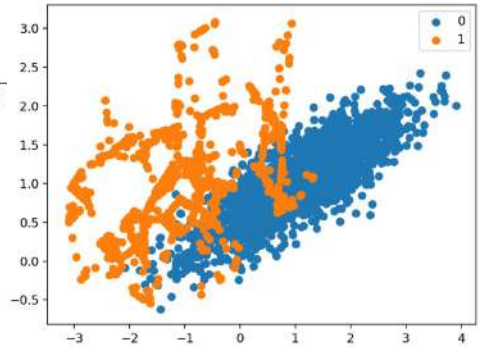
3. Synthetic Data Generation

These techniques ensure equal representation from minority groups by appropriately modifying the data sampling by

Synthetic Minority Over-sampling Technique

1. Pick a sample from the minority class.
2. Find its k nearest neighbors (default k=5)
3. Choose one randomly.
4. Create a new synthetic point somewhere





Bias Mitigation Techniques

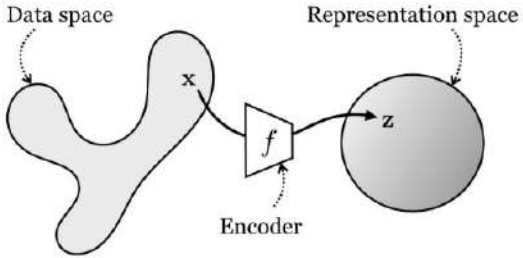
Preprocessing Techniques

Preprocessing Techniques

Modify dataset before training to remove bias

1. Fair representation learning
2. Feature modification and removal
3. Fair kernel methods

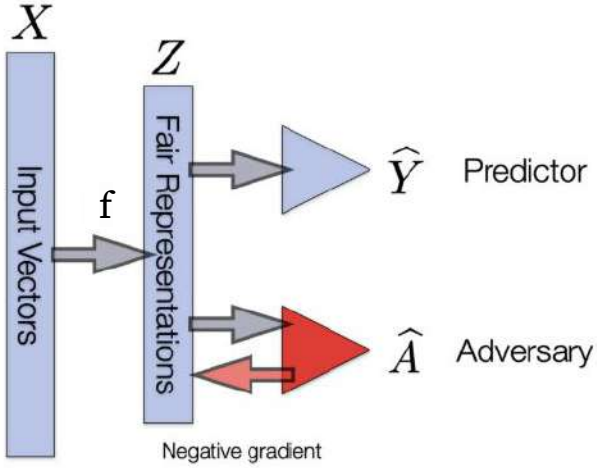
1. Fair Representation Learning



- **Learn a mapping from *X (raw data) to Z* that is informative about *Y (outcome)* but uninformative about *A (protected group)*.**
- an information-theoretic optimization problem, balancing two objectives
  - maximize mutual information between *Z* and the target *Y: I(Z;Y)* and
  - minimize mutual information between *Z* and the protected attribute A: I(Z;A).
- The goal is to learn a transformation *f: X →Z* such that:

$$max_f \;\; I(f(X);Y) - \lambda\, I(f(X);A)).$$

- Here $\lambda>0$ is a trade-off parameter that balances fairness
  - E.g., Autoencoder
  - prototype learnin



Learning Fair Representations

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https://www.cs.toronto.edu/~toni/Papers/icml-final.pdf

2. Feature Modification Approaches

Identify **protected attributes and their proxies** and remove, replace, or repair them before model training.

There are several approaches within this category:

- **Feature removal:** Simply delete features X□ that are highly correlated with A.

- **Feature transformation:** Modify features X□ to decorrelate them from A (e.g., via residualization). A common technique is one where features are regressed on A and the residuals are used: $X_j' = E[X_j|A]$.

2. Feature Modification Approaches

- **Distribution alignment**: Adjust feature distributions so that they are similar across different values of A (akin to "repair"). The repair process can be framed as minimizing the statistical distance between conditional distributions:

$$min \sum_{j=1}^{d} dist(D(X_j|A = 0), D(X_j|A = 1))$$

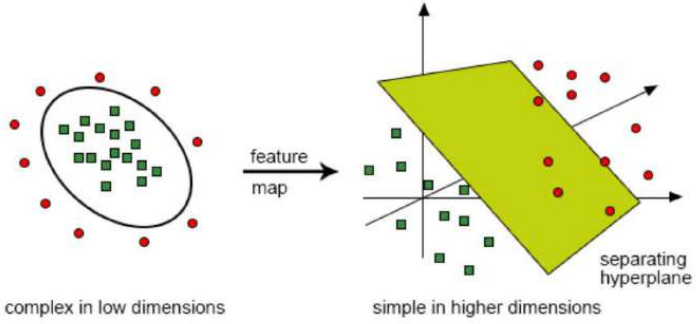
- Where *dist* is a divergence metric such as KL divergence between conditional distributions D.

$$D_{\mathrm{KL}}(P \parallel Q) = \sum_{x \in \mathcal{X}} P(x) \log \frac{P(x)}{Q(x)}.$$

Disparate Impact Remover: **Adds noise to features** so marginals match exactly. (https://dl.acm.org/doi/pdf/10.1145/2783258.2783311)

Optimal Transport Repair: **Transports mass between group distributions.** (https://proceedings.mlr.press/v97/gordaliza19a/gordaliza19a.pdf)

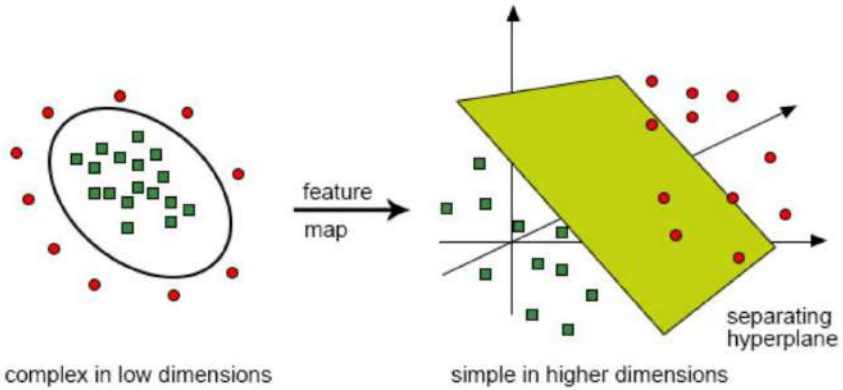
3. Kernel methods



Implicitly project input data into high-dimensional feature spaces where linear models can perform complex classification.

Learn the kernel function to ensure fairness.

3. Kernel methods



The goal is to learn a fair kernel mapping ϕ: X → H s.t. the resulting kernel k(x,x') = <ϕ(x), ϕ(x')> satisfies two properties,

1) invariance to A i.e., ϕ(x) should contain minimal information about A and

2) preservation of task-relevance i.e., ϕ(x) should be informative for predicting Y.

Bias Mitigation Techniques

In-Processing Techniques

1. Fairness-constrained optimization

The objective is to minimize a standard prediction loss (e.g., cross-entropy or MSE) while satisfying a fairness constraint.

Formalizes fairness as an explicit/implicit constraint within the optimization process:

$$min_{\theta} L_{task}(\theta) \text{ subject to } \Delta_{fairness}(\theta) \leq \epsilon .$$

- θ represents the model parameters,
- L_task(.) is the task-specific loss function (such as classification loss),
- and fairness Δ(θ) measures fairness violations (e.g., statistical parity difference or equalized odds gap).

2. Adversarial Debiasing

Two-player game approach:

“maximize the model’s ability to predict the output, while minimizing the adversary’s ability to predict the protected attribute”

The goal is to learn representations that are predictive of the task but invariant to sensitive attributes.

2. Adversarial Debiasing

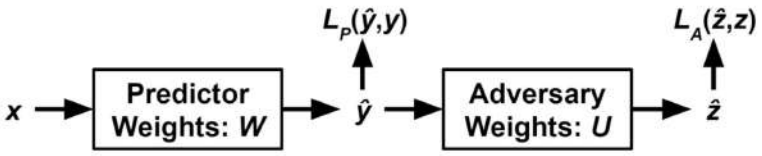


Figure 1: The architecture of the adversarial network.

Predictor Model
- learns function y = f(x)
- minimizes loss L_p(ŷ, y)

Adversary Model
- learns function z = g(y)
- minimizes loss L_a(ẑ, z)

Mitigating Unwanted Biases with Adversarial Learning

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3. Fairness-regularized optimization

Fairness is incorporated directly into the loss function as a regularization term.

This modifies the original objective to penalize fairness violations, leading to the following formulation:

$$L_{total}(\theta) = L_{task}(\theta) + \lambda L_{fairness}(A, f_{\theta}(X))$$

Bias Mitigation Techniques

Post-Processing Techniques

1. Threshold Adjustment by Group

Group-specific decision thresholds are applied to the predicted probability scores to equalize fairness metrics.

$$Y = 1 \text{ if } \Pr(Y=1|X) > t_A \text{ and } 0 \text{ otherwise.}$$

2. Re-ranking

Suppose the model assigns a score si to each individual i, the standard ranking orders individuals by descending si.

Re-ranking alters this order to ensure fair representation from different groups in the top-k list.

Fair ranking algorithm (FA*IR) that ensures k-selection fairness under specified group quotas.

FA*IR: A Fair Top-k Ranking Algorithm

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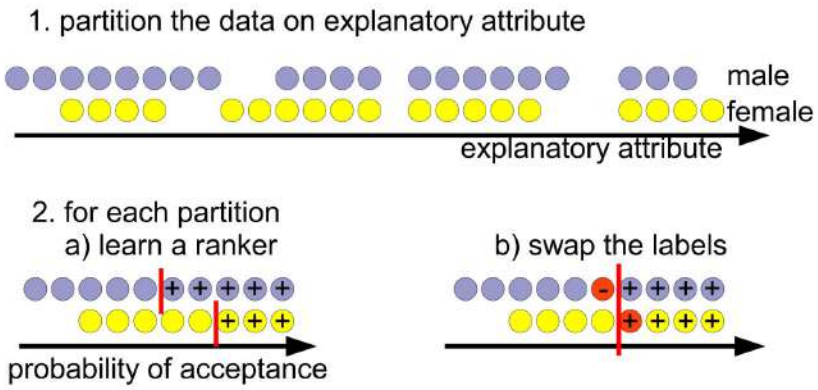
$$Skew_{group,k} = \frac{Fraction\ of\ group\ members\ in\ top\ K}{Fraction\ of\ group\ members\ overall}$$

3. Reject Option Classification

after training by altering predictions in a region of uncertainty—specifically, where the model is least confident.

In this “reject option” band, the model changes the label in favor of the disadvantaged group.

Flipping decisions in ambiguous regions.



Kamiran, F., Karim, A., and Zhang, X. Decision theory for discrimination-aware classification. In IEEE International Conference on Data Mining, pp. 924–929, 2012. doi: https://doi.org/10.1109/ICDM.2012.45.

Bias Mitigation Techniques

Beyond Statistical Metrics for Fairness in AI

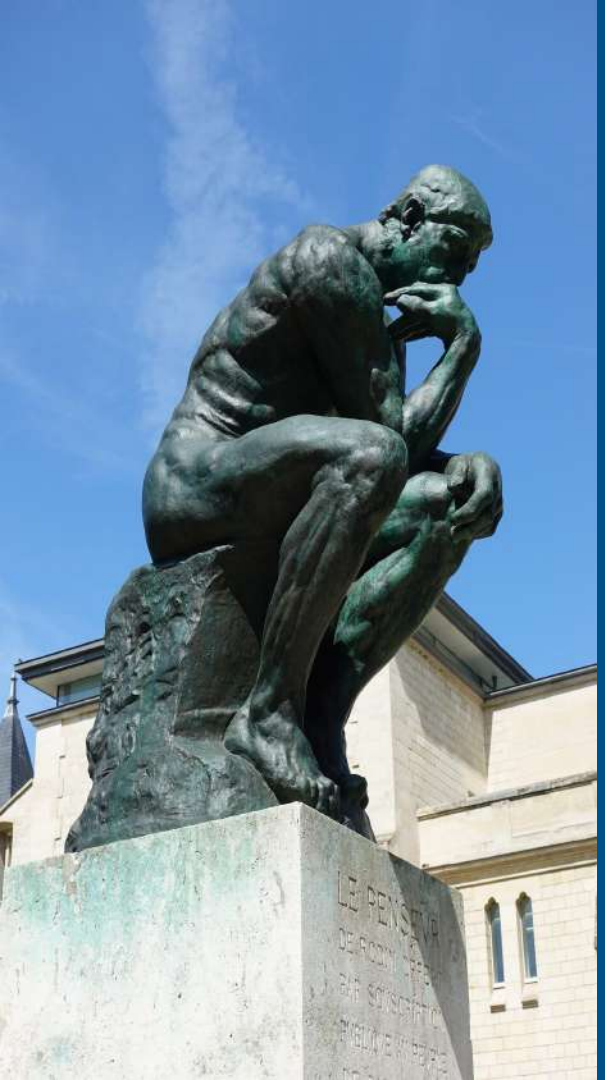
Summary of Bias Mitigation techniques

1. Data-Level Interventions (Sampling/re-weighting)
2. Preprocessing: Modify dataset before training
    a. Feature transformation/removal,
    b. learning fair representation/kernel
3. In-processing: Incorporate fairness during training
    a. modified optimization function,
    b. adding fairness constraint in loss function,
    c. incorporated fairness directly into the loss function as a regularization term
4. Post-processing: Adjust outputs after training
    a. Threshold Adjustment,
    b. Re-ranking
    c. Reject Option classification

Beyond Statistical Metrics for Fairness in AI

While statistical fairness metrics provide valuable tools for evaluating AI systems, achieving true fairness requires a broader, more holistic approach.

- Contextual understanding of historical and structural inequities
- Accountability
- Transparency and explainability
- Ethical awareness and responsibility among developers
- a combination of technical solutions, organizational change, and social responsibility.



Question: AI decides which patients get priority treatment. The patients from cities always go first, whereas the patients from villages wait longer.

What issues in the data might be causing this? How to fix it?